



Pranjal
Mechanical Engineering
Indian Institute of Technology Bombay

210100117
B.Tech.
Gender: Male
DOB: 04/11/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2025	

Pursuing a Minor Degree in **Artificial Intelligence and Data Science** under the Department of C-Minds, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Achieved AIR **1197** in **IIT JEE Advanced** | Placed in **Top 0.3 %** in JEE Mains | Conferred Scholarship of INR **1L** ['21]
- Attained a **perfect SPI** of **10** during the summer semester, securing **AA grade** across all core courses ['23]
- Aced **NDA** exam conducted by **UPSC**, showcasing exceptional proficiency in **maths** and **analytical** skills ['21]

PROFESSIONAL EXPERIENCE

Mercedes-Benz Research & Development India | *GenAI based Recommendation Engine* [May'24 - Jul'24]

- Designed and executed a proof-of-concept project, crafting a **generative AI** solution for **AIOps** by harnessing **LLMs**
- Integrated **OpenAI GPT-4o** model and **Azure AI Search** to perform **NLP** tasks and search through historical incidents
- Expanded solution's utility to incorporate the analysis of **Datadog metrics**, enabling effective **root cause analysis**
- Incorporated recommendation engine & **anomaly detection** in AIOps saving **28 human hours** of a **14-member** team

Defence Research and Development Organisation | *Quantum Cryptography Research* [May'23 - Jul'23]

Awarded **Letter of Recommendation** for exemplary performance and dedicated work during the Internship

- Utilized **Quantum Cryptography** to ensure a secure communication between two parties that is difficult to decrypt
- Created a **Simulink** model in Matlab for the **simulation** of protocol; Did a *Literature review* of 20+ research papers

TECHNICAL PROJECTS

Classifier in neurodegenerative disorders | *Proteomics lab* | *Prof. Srivastava, IIT Bombay* [Jan'24 - Apr'24]

- Employed diverse **spectroscopy** techniques to extract raw proteomics data from an authentic brain tumor sample
- Conducted **exploratory data analysis** and visualization, utilizing **PCA**, **VIF analysis**, **Correlation heatmaps**
- Utilized **machine learning** algorithms to interpret classifiers, enhancing understanding through **predictive analysis**

Architectural Layout Intelligence | *Computer Vision and ML* | *Prof. Vinay Kulkarni* [Jan'24 - Apr'24]

- Improved **architectural layout design process** through advanced classification using **optimal** past layouts
- Conducted EDA** to identify crucial layout features; **Extracted & engineered features** using **OpenCV** & **scikit-learn**
- Offered best recommendations, optimizing the starting point for new designs, significantly **reducing time** and **effort**

Strategized location of Wireless Network Devices | *Optimization* | *Prof. Avinash Bhardwaj* [Jan'24 - Apr'24]

- Formulated** and translated strategic objectives into **mathematical models** for optimizing the placement of WAPs
- Implemented **Voronoi heuristic approach** with **derivative-free optimization**, and simulated using Python coding
- Evaluated the outcomes of our implemented model against existing solutions and assessed **scalability**

DIVERSE MINI PROJECTS

- Disney+ Hotstar clone**: Created **responsive & interactive** user interface on the **front-end** that emulates the website
- Invisibility Cloak**: Extracted static background frame to create a real-time **mask** that makes objects appear **invisible**
- Obstacle Maneuvering Omni Bot**: Engineered 3-wheeled obstacle avoiding bot with **Wi-Fi** module using **Blynk IoT**
- Heart Disease Prediction**: Classified patients using Neural Networks, optimizing performance with 76 attributes
- Data Science project mentor**: Mentored a team of **4** sophomores in Winter projects, organised by **Analytics club**

TECHNICAL SKILLS

- Languages & Libraries** : C++, Python, MATLAB, Tensorflow, PyTorch, OpenCV, HTML, JavaScript, $\LaTeX$
- Applications** : PyCharm, SolidWorks, Ansys, Arduino, Blender, Eagle, Autodesk, Microsoft Office

EXTRA-CURRICULAR ACTIVITIES

- Interned at **Coding Ninjas**, contributed to **PR**, **Marketing** & **project management** from conception to completion
- Secured over **10 lakh rupees** in **funding** for the **Tinkerers' Lab**, 24/7 facility designed to empower engineers
- Featured in an **interview** on Sankalp Bharat's **YouTube** channel, which boasts over **700k+ subscribers**
- Managed a **3-week bootcamp** for **underprivileged** students, actively promoting and encouraging **STEM** education
- Led an **IT/Software** internship prep program at **IIT Bombay**, guiding over **2,000** aspiring interns towards success